#; Decimal to binary and Binary to Decimal

```
.data
  deci : .asciiz "Enter Decimal Number: "

.text
main:
  li $v0,4
  la $a0,deci
  syscall

  li $v0,5
  syscall
  move $t0,$v0

  li $t1,2  #; divisor
  li $t2,0  #; stack counter


DecimalToBinary:
   li $t1,2

   beqz $t0,printBinary

   div $t0,$t1
   mfhi $t3
   mflo $t0

   addi $sp,$sp,-4
   sw $t3 , 0($sp)
   addi $t2,$t2,1


   j DecimalToBinary

printBinary:
   beqz $t2,Exit

   lw $t3,0($sp)
   addi $sp,$sp,4
   addi $t2,$t2,-1

   li $v0,1
   move $a0,$t3
```

```
   syscall

   j printBinary

Exit:
  li $v0,10
  syscall
```

```
##################################################################################
```

#; Binary To Decimal

```
.data
   bin : .asciiz "Enter Binary Number: "
   result : .asciiz "Decimal value is: "

.text
main:
  li $v0,4
  la $a0,bin
  syscall

  li $v0,5
  syscall
  move $t0,$v0

  li $t1,10 #divisor
  li $t2,2  #base
  li $t3,0  #decimal

BToD:
   beqz $t0,print_result

   div $t0,$t1
   mfhi $t4  #;remainder
   mflo $t0  #; quatient also store $t0

   mul $t3,$t3,$t2
   add $t3,$t3,$t4

   j BToD
```

```
print_result:
   li $v0,4
   la $a0,result
   syscall

   li $v0,1
   move $a0,$t3
   syscall

   li $v0,10
   syscall
```

##

#;get factorial using stack point

```
.text
main:
   li $v0, 4
   la $a0, prompt
   syscall

   li $v0,5
   syscall
   move $a0,$v0

   jal fact

   #; print result
   move $t0,$v0
   li $v0, 4
   la $a0 , result
   syscall

   li $v0,1
   move $a0,$t0
   syscall

   li $v0,10
   syscall


fact:
   addi $sp, $sp, -8
   sw $ra,4($sp)  #;return value
```

```
  sw $a0,0($sp)  #; saved n value

  li $t0,1
  beq $a0, $t0, base

  addi $a0, $a0, -1
  jal fact

  lw $a0,0($sp)
  mul $v0,$v0,$a0
  j endf


base:
  li $v0, 1

endf:
  lw $ra,4($sp)
  addi $sp,$sp, 8
  j $ra

.data
  prompt : .asciiz "Enter Your number: "
  result : .asciiz "Factorial is:"
```

##

#; fibbonacci -> 0 1 1 2 3 5 8 using stack point

```
.data
  prompt : .asciiz "Enter your Number: "
  result : .asciiz "Fibonacci is: "

.text
main:

  li $v0,4
  la $a0,prompt
  syscall

  li $v0,5
  syscall
  move $a0,$v0
```

```
    jal fibbonacci

    move $t0, $v0          #; Save result
    li $v0, 4
    la $a0, result
    syscall

    #; Print fibonacci number
    li $v0, 1
    move $a0, $t0
    syscall

    li $v0,10
    syscall



fibbonacci:
    addi $sp,$sp,-12
    sw $ra,8($sp)
    sw $s0,4($sp)
    sw $a0,0($sp)

    beqz $a0,basezero

    li $t0, 1
    beq $a0,$t0,baseone

    addi $a0,$a0,-1
    jal fibbonacci
    move $s0,$v0

    lw $a0,0($sp)
    addi $a0,$a0,-2
    jal fibbonacci

    add $v0,$s0,$v0
    j endf

basezero:
    li $v0,0
    j endf

baseone:
```

```
  li $v0,1
  j endf

endf:
  lw $s0,4($sp)
  lw $ra,8($sp)
  addi $sp,$sp,12
  jr $ra


#################################################################################
```

#;sum of squre given number

```
.data
 N : .word 5
 sumofsqure : .word 0

.text
main:
 lw $t0,N
 li $t1,1
 li $t2,0

sum_loop:
 blt $t0,$t1,printf

 mul $t3,$t1,$t1
 add $t2,$t2,$t3

 addi $t1,$t1,1

 j sum_loop

printf:
 sw $t2,sumofsqure

 li $v0,1
 lw $a0,sumofsqure
 syscall

 li $v0,10
 syscall
```

```
#############################################################################
#;using float

.data
 text1 : .asciiz "Enter Float Number1: "
 text2 : .asciiz "Enter Float Number2: "

.text
main:
  li $v0,4
  la $a0,text1
  syscall

  li $v0,6
  syscall
  mov.s $f2,$f0

  li $v0,4
  la $a0,text2
  syscall

  li $v0,6
  syscall
  mov.s $f4,$f0

  add.s $f6,$f2,$f4

  li $v0,2
  mov.s $f12,$f6
  syscall

  li $v0,10
  syscall


#############################################################################
.data
  array  : .word 1, 2, 3, 4, 5, 6, 7, 8, 9, 10
           .word 11, 12, 13, 14, 15, 16, 17, 18, 19, 20
           .word 21, 22, 23, 24, 25, 26, 27, 28, 29, 30
  leng : .word 30

  sum : .word 0
  average : .word 0
```

```
.text
main:
  la $t0,array
  li $t1,0  #; i=1
  lw $t2,leng
  li $t3,0

sumOfloop:
  beq $t2,$t1,print_result

  lw $t4, ($t0)

  add $t3,$t3,$t4
  addi $t1,$t1,1

  add $t0,$t0,4

  j sumOfloop

print_result:
   sw $t3,sum

          li $v0,1
          move $a0,$t3
          syscall

          li $v0,10
          syscall
```